A Contradictory Taste of Competitive Mathematics

Problem 1

Prove that there do not exist integers a, b such that $a^2 - 4b = 2$.

Solution 1

Suppose there did exist such an a, b. Then, $a^2 = 2(2b+1)$ which is even so a must be even (why?). Write $a = 2A$ as integer A. Then,

$$\begin{aligned} 4A^2 &= 2(2b+1), \\ A^2 - b &= \frac{1}{2}, \end{aligned}$$

but this is a contradiction as $A^2 - b \in \mathbb{Z}$.

Note: A direct proof is to use the fact that $n^2 \equiv 0, 1 \pmod 4$.

Problem 2

Show that $\log_{10}(2)$ is irrational.

Solution 2

Suppose otherwise, that is $\dfrac{\ln 2}{\ln 10} = \log_{10}(2) = \dfrac{p}{q}$ for positive integer p, q. Then,

$$\begin{aligned} p \ln 10 &= q \ln 2, \\ 10^p &= 2^q. \end{aligned}$$

The LHS is divisible by 5 (since $p \geq 1$) but the RHS is not. This is a contradiction.

Problem 3

Prove that if a, b, c are three odd integers, then $ax^2 + bx = c$ has no rational roots.

Solution 3

Assume there is a rational root $x = \dfrac{p}{q}$. What we will be studying is parity, so the only enforcement on p, q is that they cannot both be even (why?), and that $q \neq 0$.

Substitute and cross multiply:

$$\begin{aligned} a\frac{p^2}{q^2} + b\frac{p}{q} &= c, \\ p(ap + bq) &= cq^2. \end{aligned}$$

Now, if p was even, the left hand side would be even. Thus, cq^2 and hence q is even, but this is impossible. So both p and q are odd. But then we have $ap + bq$ is even, so $p(ap + bq)$ is also even, while cq^2 is odd.

Problem 4

Show that there does not exist a function $f : \mathbb{Z} \to \{1, 2, 3\}$ satisfying $f(x) \neq f(y)$ for all $x, y \in \mathbb{Z}$ such that $|x - y| \in \{2, 3, 5\}$.

Solution 4

Assume such a function exists. Since $f(1)$ is distinct from $f(3), f(4), f(6)$, two of the three must be equal. Since $f(6)$ is distinct from $f(3)$ and $f(4)$ we must have $f(3) = f(4)$.

Since $f(7)$ is distinct from $f(5), f(4), f(2)$ the same argument applies, so $f(3) = f(4) = f(5)$. But this is a contradiction as $|5 - 3| = 2$.

Note: This is a graph theory problem in disguise and related to Ramsey Theory.

Problem 5

Prove that infinitely many primes are three more than a multiple of 4.

Solution 5

Suppose otherwise, that is there are only $n \in \mathbb{Z}^+$ primes in the form $4k + 3$. Enumerate these as $p_1 = 3, p_2, \cdots, p_n$, and consider the integer $P = 4p_2p_3 \cdots p_n + 3$. This number is not divisible by $p_1 = 3$, and also not divisible by any of $p_2, p_3, \cdots, p_n$. It cannot be a prime, or it would contradict our assumption, so it must have only prime factors in the form $4k + 1$.

But note that $(4a + 1)(4b + 1) = 4(4ab + a + b) + 1$, so a product of any amount of primes in the form of $4k + 1$ would itself be in the form $4k + 1$, and cannot be P.

Problem 6

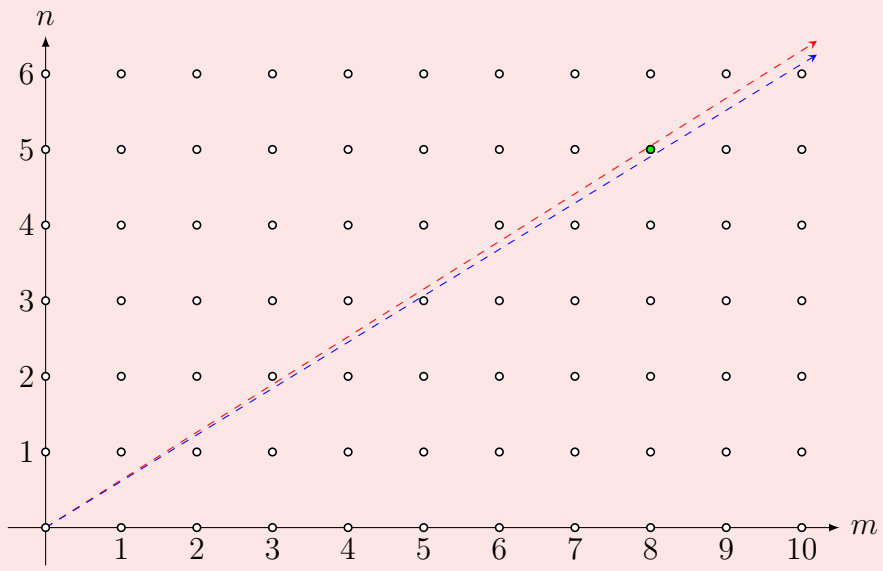
Show that there does not exist a strictly increasing function $f : \mathbb{Z}^+ \to \mathbb{Z}^+$ satisfying $f(2) = 3$ and $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{Z}^+$.

Solution 6

Suppose such a function existed. It is apparent that $f(2^m) = f(2)^m = 3^m$. Now, since f is increasing, we have $f(3) < f(4) = 9$, and $f(3)^2 = f(9) > f(2^3) = 27$, so $f(3) > 5$. So $f(3) = k$, where $k \in \{6, 7, 8\}$, and $f(3^n) = f(3)^n = k^n$.

The reason we have done this is to study how fast the two "lineages" $f(2^m)$ and $f(3^n)$ grow, with respect to their inputs. Note that if $3^n > 2^m$, we must have $f(3^n) > f(2^m)$. The first condition occurs when $n > m \log_3 2$, while the second condition occurs when $n > m \log_k 3$. But observe that $\log_k 3 < \log_3 2$ for all our possible k, so intuitively there will eventually be m, n where we can satisfy $f(3^n) > f(2^m)$ but not $3^n > 2^m$, a contradiction.

A way to visualise this is to plot the inequalities on the same pair of axis. For $k = 6$:



The green lattice point lies between the red ($n = m \log_3 2$) and blue ($n = m \log_6 3$) lines. So we would expect it to provide us a contradiction. Indeed, for $(m, n) = (8, 5)$,

$$3^5 = 243 < 256 = 2^8,$$
$$f(3^5) = 6^5 = 7776 > 6561 = 3^8 = f(2^8).$$

In fact, this pair of m, n works as a contradiction to all possible values of k (why?).

Note: This proof can be modified to show no function exists if $\log_2 f(2) \notin \mathbb{Z}^+$.

Problem 7

Given any sequence of $mn + 1$ real numbers, show that some subsequence of length $(m + 1)$ is increasing or some subsequence of length $(n + 1)$ is decreasing.

Solution 7

Notate our sequence $a_i, a_2, \ldots a_{mn+1}$. For each $1 \leq i \leq nm + 1$ look at ℓ_i, the length of the longest increasing subsequence ending at a_i of the first i element of our sequence.
We now have two cases:

- For at least one i, we have that $\ell_i \geq m + 1$. In that case, form definition we are done.
- We have $\ell_i \leq m$. Then $\ell_i \in \{1, 2, \ldots m\}$ for all i. By the pigeonhole principle on our sequence of length $mn + 1$, at least one value in $\{1, 2, \ldots m\}$ occurs at least $n + 1$ times. Suppose these i correspond to $a_{i_1}, a_{i_2}, \ldots a_{i_{n+1}}$. Then there must form a strictly decreasing subsequence else to contradict our definition of ℓ_i.

Note: this is the Erdős–Szekeres theorem and can also be proven using Dilworth's

theorem. Further, this proof is similar to the ones that shows that every sequence in $\mathbb{R}$ contains a monotone sub-sequence (see here).

Problem 8

Every point of three-dimensional space is coloured red, green or blue. Show that one of the colours attains all distances; that is, any positive real number represents the distance between two points of this colour.

Solution 8

If we only used one colour to colour the whole space, the problem is trivial. The case where we only use two colours is left to the reader. So assume we use all three colours, and yet no such colouring exists. This would require there to be a minimum distance $r, g, b > 0$ for which no two red, green, blue points are separated by, respectively, in any three colouring of the space. Since the colours are arbitrary, assume that $r \leq g \leq b$.

Consider some blue point, and construct a sphere of radius b around it. This sphere cannot have any blue points on it (why?). If it had no green points on it, it is entirely red, but then we could find two red points with a separation of r (as $r \leq b$, convince yourself this in fact holds for $r \leq 2b$). So there is some green point on the sphere.

Draw a sphere with radius g on this green point, as $g \leq b$ it follows that our two spheres intersect, and the intersection is a circle $\mathcal{C}$. In the case when $g = b$, the cross section of a plane through the centres of the spheres would look like the classic construction of an equilateral triangle with compass and straightedge, and is the case where the ratio of the radius of $\mathcal{C}$ to g is the smallest (why?), being $\sin(60°) = \sqrt{3}/2$. That is, the diameter of $\mathcal{C} \geq \sqrt{3}g > g$, and because all the points on this circle have to be red (why?), it is always possible to find two red on this circle with separation $r \leq g$, a contradiction. This is essentially the proof you need for the two colouring case.

Problem 9

Show that no set of nine consecutive integers can be partitioned into two sets with equal product of elements.

Solution 9

Suppose there does exist some integer n for which the set $\{n, n+1, \cdots, n+8\}$ can be partitioned into sets A and B which satisfy the requirement. If $n \in \{-8, -7, \cdots, 0\}$, exactly one of the sets will have 0 in it, while the other will have nonzero product of elements. Otherwise, if $n < -8$, all of the numbers $\{n, n+1, \cdots, n+8\}$ will be negative, so since the parity of the amount of elements in each set is different, one product will be positive and the other negative. These are both contradictions, so $n \geq 1$.

One of A or B has at most 4 elements, and hence has product at most $p_1 \leq (n+5)(n+6)(n+7)(n+8)$. The other one hence must have product at least $p_2 \geq n(n+1)(n+2)(n+3)(n+4)$. Then, since $p_1 = p_2$, we have the inequality

$$n(n+1)(n+2)(n+3)(n+4) \leq (n+5)(n+6)(n+7)(n+8),$$

already we can see that we will only have to check a finite amount of cases because the LHS is a polynomial of a higher degree and will surely grow past the RHS.

Checking by hand, $n = 6$ does not satisfy the inequality, and so we verify by hand that for $n = 1, 2, 3, 4, 5$, $\sqrt{n(n+1)\cdots(n+8)} = \sqrt{p_1 p_2} = p_1$ is never an integer. It never is, so we have a contradiction. A rigorous proof that $n < 6$ can be achieved by expanding the polynomials and differentiating.

Note: This is an immediate consequence of a theorem of Erdős.